

The propositions package*

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Abstract

The `propositions` package provides a key-value driven system for labelling propositions, theses, and premises in academic papers. Items may be given names like ‘(P)’ or ‘Physicalism’, or auto-numbered using different counters; all carry robust cross-references with configurable formatting. The package integrates with `amsmath`, `hyperref`, and `cleveref`.

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1 Introduction

In some academic disciplines (such as philosophy), it is common to have displayed propositions (examples, theses, premises, . . .) with various kinds of labels. A thesis might be referred to as ‘(P)’ or ‘Physicalism’; the premises of an argument might be numbered as ‘P1’, ‘P2’, ‘P3’, . . . ; or examples might be numbered consecutively over the course of a whole article. Standard L^AT_EX environments like `enumerate` can handle some of these cases, but cross-referencing is awkward: `\ref` produces a bare number or letter, and the author must manually add parentheses or other formatting at every point of reference. The standard `description` environment does not allow cross-referencing at all.

The `propositions` package solves this by attaching formatting information to each label. A short item like `\item[P]` (inside `prop`) is displayed as “(P)” and `\ref` automatically produces “(P)” as well—complete with parentheses, hyperlinks, and `cleveref` support. The full key-value interface supports named items, numbered items, custom counters, glosses, shorthands, and per-item format overrides.

The `\ptag` command (which requires `amsmath`) extends this to displayed math environments: an equation can be tagged with a proposition label instead of (or using) its equation number.

2 History

I wrote the ancestor to this package in the 90s while finishing my Ph.D. thesis, but never documented it or shared it with the world. This new version is a thorough re-implementation in L^AT_EX3, written in 2026 with extensive help from Claude Code. I hope others will find it as useful as I have.

3 Basic usage

Load the package with `\usepackage{propositions}` or `\usepackage[options]{propositions}` (see [section 9](#) below for valid package options).

The `prop` environment generates a list of propositions, each introduced by `\item`. `\item` with an optional argument gives a `description`-like label:

<pre>\begin{prop} \item[Physicalism] Everything is physical. \label{phys} \item[Idealism] Everything is mental. \label{ideal} \end{prop}</pre>	<pre>Physicalism Everything is physi- cal. Idealism Everything is mental.</pre>
--	---

Unlike the standard `description` environment, one can refer back to these propositions using the standard `\ref` command:

<pre>\ref{phys} is more plausible than \ref{ideal}.</pre>	<pre>Physicalism is more plausible than Idealism.</pre>
---	---

With no optional argument, `\item` will by default generate numbered items similar to `enumerate`, but with numbering that persists across the document:

<pre> \begin{prop} \item Every atom is physical. \label{atoms} \end{prop} \ref{phys} follows from the conjunction of \ref{atoms} and \begin{prop} \item Everything is an atom. \label{atomism} \end{prop} </pre>	(1) Every atom is physical. Physicalism follows from the conjunction of (1) and (2) Everything is an atom.
--	--

As with `enumerate`, the counter and formatting depend on the nesting level:

<pre> \begin{prop} \item \label{dual} \begin{prop} \item Some things are physical. \label{some} \item Some things are not physical. \label{notall} \end{prop} \end{prop} \end{prop} Without \ref{some}, \ref{dual} would be consistent with \ref{ideal}. </pre>	(3) a. Some things are physical. b. Some things are not physical. Without (3a), (3) would be consistent with Idealism.
---	---

4 Advanced usage

The format of the proposition labels, and of subsequent references, are both configurable using a key=value syntax (see [section 6](#) for the possible keys):

<pre> \begin{prop} \item[No Overlap, align=flush, display format=\textsc{#1}, ref format=\textit{#1}] Nothing mental is physical. \label{incomp} \end{prop} Is \ref{incomp} consistent with \ref{phys}? </pre>	No OVERLAP Nothing mental is physical. Is <i>No Overlap</i> consistent with Physicalism?
--	---

Preset *prop styles* can be declared and used instead of configuring the `\items` one at a time:

<pre> \begin{prop} \item[Nihilism, style=thesis] There is nothing. \label{nihilism} \end{prop} Does \ref{nihilism} imply \ref{phys}, \ref{dual}, or both? Discuss. </pre>	NIHILISM There is nothing. Does NIHILISM imply Physicalism, (3), or both? Discuss.
---	---

Shortcut commands—e.g., `\titem[⟨keys⟩]` for `\item[style=thesis, ⟨keys⟩]` (within `prop`)—can also be defined. The package loads with a range of predefined styles.

When the package is loaded with `\usepackage[equations]{propositions}`, first-level `\items` within `prop` use the same counter as equations. (This looks better with the `leqno` option to `\documentclass`.)

<pre>\begin{equation} \exists x (\text{Mental}(x) \wedge \text{Physical}(x)) \end{equation} \begin{prop} \item There is overlap between the mental and the physical. \end{prop}</pre>	$(4) \quad \exists x(\text{Mental}(x) \wedge \text{Physical}(x))$ (5) There is overlap between the mental and the physical.
---	--

The `\ptag` command (requires `amsmath`) is an analogue of `\item` (within `prop`) that works inside displayed math environments.

<pre>\begin{equation} \ptag{Monism} \label{mon} \exists x \forall y (y = x) \end{equation} Is \ref{mon} compatible with \ref{dual}?</pre>	$\text{Monism} \quad \exists x \forall y (y = x)$ Is Monism compatible with (3)?
---	--

5 The `prop` environment and `\pitem`

```
\begin{prop}[⟨keys⟩]
  ⟨environment content⟩
\end{prop}
```

Creates a displayed list of propositions. It is a standard L^AT_EX list, so by default its formatting will depend on the standard length parameters like `\itemsep` and `\topsep`, although these can be overridden by setting keys.

Within `prop`, `\pitem` (see below) creates labelled items. The ordinary `\item` command is still available for unlabelled items.

Any display math environment (`\equation`, `align`, etc.) used inside `prop` or `inlineprop` automatically increments the nesting level for its duration, so that `\ptag` inside such an environment behaves as if it were one level deeper. The optional `⟨keys⟩` argument accepts the same keys as `\proppositions`^{P. 15} (section 6), with effects local to this environment.

```
\begin{inlineprop}[⟨keys⟩]
  ⟨environment content⟩
\end{inlineprop}
```

Like `prop`, but does not create a list. Allows `\pitem` to be used outside list environments, e.g. for generating numbers at the beginning of paragraphs. Steps the `prop` counter and increments the nesting level. Accepts the same optional `⟨keys⟩` as `prop`.

Within `prop` and `inlineprop`, `\item` is redefined to act as the proposition-item command. Its optional argument is a comma-separated list of $\langle key \rangle = \langle value \rangle$ pairs (see [section 6](#)). A bare string without `=` is treated as a proposition name.

When used without an optional argument (or without setting `name`, `counter`, or `style`), the style is determined by the `nameless style` key ([section 6](#)). By default this is `numbered`, which dispatches by nesting depth to `levelone`, `leveltwo`, ... via `\proplevelchoice`.

`\pitem[ $\langle keys \rangle$ ]` (deprecated)

Deprecated alias for `\item` within `prop` or `inlineprop`. Issues a warning on each use. Replace with `\item`.

`\ptag[ $\langle keys \rangle$ ]`

(Available only when `amsmath` is loaded.) Works inside displayed math environments like `equation` and `align`. Accepts the same keys as `\item` within `prop`, except that `align` has no effect (since positioning is controlled by the tag placement system).

6 Keys

Most keys are available in three contexts:

- In the optional argument of `\item` or `\ptag`: the key applies to that item only.
- In the optional argument of `prop` or `inlineprop`: list-dimension keys ([subsection 6.2](#)) take effect immediately for the whole list; other keys act as per-item defaults, applied before each item's own explicit keys.
- Via `\propoptions`^{→P.15} or as a `\usepackage` option: the key applies globally.

Keys in [subsection 6.1](#) are meaningful only at item level. The `equations` key is global only ([subsection 6.5](#)).

6.1 Item-level keys

`name= $\langle text \rangle$` (no default)

The proposition's name. A bare string (without `=`) in the key list is equivalent to `name= $\langle text \rangle$` .

`style= $\langle style \rangle$` (no default)

A prop style, equivalent to a preset collection of keys. Styles can be declared using `\SetPropStyle`^{→P.9} or `\DeclareNumberedStyle`^{→P.10}, and several come predefined ([section 7](#)).

`counter= $\langle name \rangle$` (no default)

Counter to use. The counter is automatically stepped, and the item's name is set to `\the $\langle name \rangle$` (though this can be overridden by `counter format` or `name`).

counter format= $\langle template \rangle$ (no default)
How to display the counter value. Use #1 for the counter name, e.g. **counter format**=`\roman{#1}`. When set, this is used instead of `\the\counter` for generating the item's name.

format= $\langle template \rangle$ (no default)
Formatting applied to the **name**: use #1 for the argument, e.g. **format**=`\textbf{#1}`. Shorthand for setting both **display format** and **ref format**.

display format= $\langle template \rangle$ (no default)
Format for displaying the name in the proposition's label. Does not affect cross-references.

label format= $\langle template \rangle$ (no default)
Format applied to the *entire* assembled label, i.e. the name (after **display format**), shorthand, and gloss together. Use #1 for the whole assembled text. For example, **label format**=#1: appends a colon after the complete label. Does not affect cross-references.

ref format= $\langle template \rangle$ (no default)
Format for subsequent cross-references to this proposition. Does not affect the display.

shorthand= $\langle text \rangle$ (no default)
An abbreviation displayed after the name. If present, the shorthand becomes the reference text: `\ref` produces the shorthand (formatted with **ref format**) rather than the full name.

shorthand format= $\langle template \rangle$ (initially `\sim[#1]`)
Format for displaying the shorthand in the label.

gloss= $\langle text \rangle$ (no default)
A parenthetical gloss displayed after the name. Does not affect cross-references.

gloss format= $\langle template \rangle$ (initially `\sim(#1)`)
Format for displaying the gloss in the label.

ref= $\langle text \rangle$ (no default)
Explicitly set the reference text, overriding what would be derived from **name**, **counter**, or **shorthand**.

label= $\langle label \rangle$ (no default)
Equivalent to a trailing `\label{\langle label \rangle}`.

crefname= $\langle type \rangle$ (no default)
When `cleveref` is loaded, assigns an arbitrary reference type to this proposition. For example, **crefname**=`lemma` causes `\cref` to use the names defined by `\crefname{lemma}{...}{...}` instead of the default `proposition` type. The $\langle type \rangle$ must be known to `cleveref`; new types can be declared with `\crefname`.

`reset`= $\langle$ *boolean* $\rangle$ (initially `true`)

When `true` (the default), processing this `\pitem` resets the sub-level counter one nesting depth below the current level (e.g. `numpropii` at level 1). Set `reset=false` to suppress this reset, so that sub-item numbering continues from where it left off across consecutive parent items. Can also be set in a style definition via `\SetPropStyle`^{P.9}.

`align`= $\langle$ *value* $\rangle$ (no default)

How the label should be positioned. Possible values: `left` (standard left-aligned label, offset controlled by `\labelwidth` and `\labelsep`), `right` (right-aligned, like `enumerate`), `center` (centered within the label box), `flush` (aligned with left margin of item text), `nextline` (label on its own line), and `flush-nextline`. Has no effect inside `inlineprop` or `\ptag`. For left-margin alignment, use `labelindent=0em` instead.

6.2 List dimensions

These keys are effective in all three contexts. At item level, a dimension key applies only to the label of that specific item; at environment and global level it applies to the whole list.

`topsep`= $\langle$ *length* / *length list* $\rangle$
`partopsep`= $\langle$ *length* / *length list* $\rangle$
`itemsep`= $\langle$ *length* / *length list* $\rangle$
`parsep`= $\langle$ *length* / *length list* $\rangle$
`leftmargin`= $\langle$ *length* / *length list* $\rangle$
`rightmargin`= $\langle$ *length* / *length list* $\rangle$
`labelwidth`= $\langle$ *length* / *length list* $\rangle$
`labelsep`= $\langle$ *length* / *length list* $\rangle$
`itemindent`= $\langle$ *length* / *length list* $\rangle$
`listparindent`= $\langle$ *length* / *length list* $\rangle$

Override the standard L^AT_EX list dimensions. Accept the same values as `\setlength`, including rubber lengths (e.g. `itemsep=4pt plus 2pt`). Each key may also take a comma-separated list of per-level values: e.g. `leftmargin={2.5em, 0em}`. The first value applies at level 1, the second at level 2, etc. Gaps in the list (e.g. `leftmargin={, 0em}`) cause the class default to be used for that level.

`labelindent`= $\langle$ *length* / *length list* $\rangle$ (no default)

Positions the left edge of the label box at $\langle$ *length* $\rangle$ from the enclosing margin, adjusting `\labelwidth`, `\leftmargin`, `\labelsep`, or `\itemindent` as needed. (Since the value of any one of these dimensions can be derived from those of the other four, it never makes sense to set all five.)

`tightspacing` (no value)

Sets all vertical spacing to the compact defaults that the standard document classes use for level-three lists (`topsep` and `itemsep` to 2pt with stretch/shrink, `parsep` to 0pt, `partopsep` to 1pt). Individual dimension keys set afterward override.

`nosep` (no value)

Sets `\topsep`, `\itemsep`, and `\parsep` all to zero.

`continue`= $\langle\textit{boolean}\rangle$ (initially `true`)

When `true`, a `prop` environment that immediately follows another `prop` (with no intervening paragraph text) replaces the normal `\topsep`-based inter-list space with `\itemsep` + `\parsep`, giving the visual appearance of a single continuous list.

`items`= $\langle n \rangle$ (initially 1)

Indicates that approximately $\langle n \rangle$ items are expected in this environment. Used together with `\propformlabel` to compute a preview label wide enough for the $\langle n \rangle$ th expected item, so that dimension expressions such as `leftmargin=\widthof{\propformlabel}+0.5em` reserve enough space for the widest label. Has no effect on actual item processing or counter stepping.

`\propformlabel`

Expands to the formatted label of the (approximately) `items`-th item in this environment, computed at `\begin{prop}` *before* the list dimensions are applied, so it can be used directly in dimension expressions. Updated after each `\item` to reflect the most recently output label. Typical use:

```
% \begin{prop}[items=20, leftmargin=\widthof{\propformlabel}+0.5em]
%
```

sizes the left margin to accommodate labels up to the 20th item.

6.3 Default styles

When `\pitem` or `\ptag` is used without an explicit `style`, one of the following defaults is selected based on whether a name/counter was given and whether `ptag` mode is active.

`named style`= $\langle\textit{style}\rangle$ (initially `proposition`)

The style used when a `name` or `counter` is given but no explicit `style`.

`named ptag style`= $\langle\textit{style}\rangle$ (initially empty)

If set, overrides `named style` for `\ptag` items only.

`nameless style`= $\langle\textit{style}\rangle$ (initially `numbered`)

The style used when `\pitem` or `\ptag` has no optional argument (no name, counter, or style).

`nameless ptag style`= $\langle\textit{style}\rangle$ (initially empty)

If set, overrides `nameless style` for `\ptag` items only.

6.4 Environment-level style and item defaults

`style`= $\langle\textit{style}\rangle$ (no default)

Sets a default style for all `\items` in this environment. The style's list-dimension keys (`leftmargin`, `labelwidth`, etc.) take effect immediately for the whole list; item-level keys are applied as defaults for each `\item`. An explicit `style`= on an individual `\item` overrides this default.

6.5 Equation numbering

equations (no value, global only)

Sets `nameless style=eqnum` and installs hooks so that the $\text{\LaTeX}$ and `amstex` displayed equation environments (such as `equation` and `align`) use the same formatting as the `equation` prop style. The `equation` style's `display format` and `ref format` keys are used for generating equation numbers and cross-references to them. Redefining the `equation` style, e.g. with

```
\SetPropStyle{equation}{format = [#1]}
```

will automatically update the hooks.

7 Prop styles

\SetPropStyle{*<name>*}{*<keys>*}

Defines or modifies a prop style for use with the `style` key. All `\pitem`^{P.5} keys are accepted, plus the following:

style=*<parent>* (no default)

Inherit from a parent style. When an item is created, the parent's settings are loaded first (recursively, if the parent itself has a parent), then this style's own keys are applied on top.

The argument may be a macro which is expanded when the item is created: for example, a style with `style=\{<style1>\},\{<style2>\},\{<style3>\}` will resolve to `\{<style1>\}, \{<style2>\}, \{<style2>\}` depending on the nesting depth. The built-in `numbered` and `eqnum` styles use this mechanism.

macro=*<command>* (no default)

A new user macro, equivalent to `\pitem[style=<name>]`. Any further keys given to the macro are passed to `\pitem`.

If the style *<name>* already exists, `\SetPropStyle` modifies or adds keys. For example, `\SetPropStyle{proposition}{align=flush}` changes the alignment of the built-in `proposition` style while preserving its other settings.

Styles can also be created with a `pgfkeys`-like syntax via `\propoptions`^{P.15}: `\propoptions{<name>/ .style={<keys>}}` is equivalent to `\SetPropStyle{<name>}{<keys>}`.

<pre>\SetPropStyle{angle}{ labelindent = 0em, display format = \textbf{\$\langle\angle\rangle\$}, ref format = \$\langle\angle\rangle\$, macro = \angitem } \begin{prop} \angitem[Angle thesis] Everything is angular. \end{prop} No further discussion of \lastref{} is needed.</pre>	⟨Angle thesis⟩ Everything is angular. No further discussion of ⟨Angle thesis⟩ is needed.
--	---

\DeclareNumberedStyle{*<name>*}[*</keys>*]

Creates a new L^AT_EX counter named *<name>* and a matching prop style with `counter=<name>`. All `\SetPropStyle`^{→P.9} keys are accepted, plus:

parent=*<counter>* (no default)

A parent counter; the new counter resets when the parent steps (same mechanism as `\numberwithin`). A dedicated **prop** counter (stepped by each **prop** and **inlineprop**) is available for non-persistent numbering.

<code>\DeclareNumberedStyle{P}</code> <code>\begin{prop}</code> <code> \item[counter=P] First premise.</code> <code> \label{p1}</code> <code> \item[counter=P] Second premise.</code> <code> \label{p2}</code> <code>\end{prop}</code> <code>From \ref{p1} and \ref{p2}\ldots</code>	P1 First premise. P2 Second premise. From P1 and P2 ...
---	---

\proplevelchoice{*<item1, item2, ...>*}

Picks the entry at the current nesting depth (1-indexed, clamped to the list length). At depth 0 (outside any **prop** environment), returns the first entry. This macro is fully expandable and can be used in the **name**, **format**, and **style** keys of `\SetPropStyle` to create depth-dependent styles.

7.1 Built-in styles

The following prop styles are predefined. Each entry shows the defining code (using user-facing commands) and a live example. All styles can be modified with `\SetPropStyle`^{→P.9}.

Text styles

plain Unformatted text label.

```
| \SetPropStyle{plain}{format = #1}
```

Claim An unformatted item.

proposition Bold text label. Auto-selected when `\item` has a name but no **style**.

```
| \SetPropStyle{proposition}{format = \textbf{#1}}
```

Thesis A bold-formatted item.

thesis Small-caps name at the text margin (flush alignment). Shortcut: `\titem`.

```
| \SetPropStyle{thesis}{format = \textsc{#1},
|   align = flush, macro = \titem}
```

HYPOTHESIS A flush-aligned item.

vignette Italic name at the text margin with a colon in the label.

```
| \SetPropStyle{vignette}{ref format = \textit{#1},  
    align = flush, label format = \textit{#1:}}
```

Example: A vignette item.

bullet Bullet symbol varying by depth (like `\itemize`). Shortcut: `\bitem`.

```
| \SetPropStyle{bullet}{align = center,  
    name = \proplevelchoice{\textbullet,  
        {\normalfont\textendash}, \textasteriskcentered,  
        \textperiodcentered},  
    display format = #1, macro = \bitem}
```

- A bullet item.

Numbered styles with dedicated counters

roman Roman-numeral counter, resets per section. Shortcut: `\ritem`.

```
| \DeclareNumberedStyle{roman}[parent = section,  
    counter format = \roman{#1}, format = (#1), macro = \ritem]
```

- (i) A roman-numbered item.

alph Alphabetic counter, resets per section. Shortcut: `\aitem`.

```
| \DeclareNumberedStyle{alph}[parent = section,  
    counter format = \alph{#1}, format = (#1), macro = \aitem]
```

- (a) An alphabetic item.

Level-specific styles

These styles are used by the **numbered** and **eqnum** dispatcher styles (below). They share the counters `numpropi` – `numpropv`, which reset automatically when an `\item` at the next lower level is processed.

levelone Level-1 numbered items: arabic in parentheses.

```
| \SetPropStyle{levelone}{counter = numpropi, format = (#1)}
```

leveltwo Level-2: lowercase alphabetic with dot.

```
| \SetPropStyle{leveltwo}{counter = numpropii,  
    display format = #1., ref format = \parentref{#1}}
```

`levelthree` Level-3: lowercase roman in parentheses.

```
| \SetPropStyle{levelthree}{counter = numpropiii,  
|   display format = (#1), ref format = \parentref{.#1}}
```

`levelfour` Level-4: uppercase alphabetic with dot.

```
| \SetPropStyle{levelfour}{counter = numpropiv,  
|   display format = #1., ref format = \parentref{#1}}
```

`levelfive` Level-5: uppercase roman in parentheses.

```
| \SetPropStyle{levelfive}{counter = numpropv,  
|   display format = (#1), ref format = \parentref{.#1}}
```

`equation` Uses the `equation` counter. When the `equations` option is active, `\SetPropStyle{equation}{...}` automatically updates the `\tagform@` and `\ref` hooks for all equations.

```
| \SetPropStyle{equation}{counter = equation, format = (#1)}
```

Dispatcher styles

These styles use `style` with `\proplevelchoice` to dispatch to a level-specific style at resolution time.

`numbered` The default nameless style. Dispatches to `levelone`–`levelfive` by nesting depth.

```
| \SetPropStyle{numbered}{style = \proplevelchoice{  
|   levelone, leveltwo, levelthree, levelfour, levelfive}}
```

`eqnum` Like `numbered`, but uses `equation` at level 1. Activated by the `equations` package option (which sets `nameless style = eqnum`).

```
| \SetPropStyle{eqnum}{style = \proplevelchoice{  
|   equation, leveltwo, levelthree, levelfour, levelfive}}
```

`hierarchical` Produces 1, 1.1, 1.1.1... numbering using `\parentref` and `counter` format. Defined via two helper styles:

```
| \SetPropStyle{h-base}{counter = numpropi,  
|   counter format = \arabic{#1}, format = #1}  
| \SetPropStyle{h-sub}{  
|   counter = \proplevelchoice{numpropi, numpropii,  
|     numpropiii, numpropiv, numpropv},  
|   counter format = \arabic{#1},  
|   format = \parentref{.#1}}  
| \SetPropStyle{hierarchical}{style = \proplevelchoice{  
|   h-base, h-sub}}
```

8 Cross-referencing

Labels placed after `\item` items (within `\prop`) work with the standard `\label/\ref` mechanism. The key difference from ordinary $\text{\LaTeX}$ references is that `\ref` produces *formatted* output: for example, `\textbf` might be applied to the name, or the number might be wrapped in parentheses. The formatting is controlled by the `format` key (or separately by `display format` and `ref format`).

`\Ref{\label}`
`\Ref*{\label}`

Titlecase variants of `\ref` and `\ref*`: uppercases the first letter of the formatted output. (For example, if `\ref{thesis}` produces ‘the Identity Theory’, then `\Ref{thesis}` produces ‘The Identity Theory’.) The starred form suppresses the hyperlink. Note: `\Ref` only affects references produced by this package (which are stored via `\propapply`); for other references, it behaves like `\ref`.

`\nref{\label}`
`\nref*{\label}`
`\Nref{\label}`
`\Nref*{\label}`

“Naked ref.” Outputs the bare reference content with all formatting stripped. If `\ref{premise}` produces ‘(P1)’, then `\nref{premise}` produces ‘P1’. The starred form suppresses the hyperlink. `\Nref` is the titlecase variant: it uppercases the first letter of the bare content.

`\nref` can be useful in the argument of `\item`, when the the name of one proposition should depend on that of another:

<code>\SetPropStyle{proposition}{format=(#1)}</code>	
<code>\begin{prop}</code>	(Phys) Everything is physical.
<code>\item[Phys]</code>	(Phys*) Almost everything is physical.
Everything is physical.	ical.
<code>\label{phys2}</code>	
<code>\item[\nref{phys2}*]</code>	((Phys*)) This one has two sets
Almost everything is physical.	of parentheses, which is probably
<code>\label{newphys2}</code>	not desired. (Phys*) avoided this
<code>\item[\ref{phys2}*]</code>	by using <code>\nref</code> .
This one has two sets of parentheses,	
which is probably not desired.	
<code>\ref{newphys2}</code> avoided this by using	
<code> \nref </code> .	
<code>\end{prop}</code>	

Warning: documents where the name of one item includes a reference to that of another, and there are further references to that item, will require multiple $\text{\LaTeX}$ runs to resolve all references. To save time, it is better to avoid long chains of dependencies of this sort.

`\oref[⟨prefix⟩][⟨suffix⟩]{⟨label⟩}`
`\oref*[⟨prefix⟩][⟨suffix⟩]{⟨label⟩}`
`\Oref[⟨prefix⟩][⟨suffix⟩]{⟨label⟩}`
`\Oref*[⟨prefix⟩][⟨suffix⟩]{⟨label⟩}`

“Ref with options.” Extends `\ref` by injecting a prefix and/or suffix *inside* the formatting. With one optional argument, $\langle suffix \rangle$ is appended; with two, $\langle prefix \rangle$ is prepended and $\langle suffix \rangle$ appended. For instance, if `\ref{premise}` produces ‘(P1)’, then `\oref[*]{premise}` produces ‘(P1*)’ and `\oref[old~][*]{premise}` produces ‘(old~P1*)’.

`\Oref` is the titlecase variant; the starred forms suppress the hyperlink.

`\oref` can also be useful in the name of `\items`, if one wants the display format for the modified item to depend on that originally used

<pre>\begin{prop} \label{premise} \item[style=plain,name=\oref[*]{phys2}] This will use boldface and parentheses because the original referenced item did. \end{prop}</pre>	<pre>(Phys*) This will use boldface and parentheses because the original referenced item did.</pre>
---	---

Another handy use for `\oref` is in combination with `\nref` to refer to ranges:

<pre>The first two numbered examples in this document were were \oref[--\nref{atomism}]{atoms}.</pre>	<pre>The first two numbered examples in this document were were (1-2).</pre>
---	--

`\lastref` [$\langle prefix \rangle$] { $\langle suffix \rangle$ }
`\Lastref` [$\langle prefix \rangle$] { $\langle suffix \rangle$ }

Produces a reference (with no hyperlink) to the most recently processed `\pitem` or `\ptag`, even without a `\label`. Useful for back-references in running text. With one argument, $\langle suffix \rangle$ is appended; with two, $\langle prefix \rangle$ is also prepended. Use `\lastref{}` for a plain reference. `\Lastref` is the titlecase variant.

`\nlastref`
`\nLastref`

Like `\lastref{}`, but returns the bare content without formatting. `\nLastref` is the titlecase variant.

`\parentref` [$\langle prefix \rangle$] { $\langle suffix \rangle$ }
`\Parentref` [$\langle prefix \rangle$] { $\langle suffix \rangle$ }

Inside a nested `prop` (or `inlineprop`), produces a formatted reference to the most recent item of the enclosing level. Same argument convention as `\lastref`. `\Parentref` is the titlecase variant.

`\nparentref`
`\nParentref`

Like `\parentref{}` but returns the bare content without formatting. Takes no arguments; simply output any desired suffix directly afterwards. `\nParentref` is the titlecase variant.

`\parentref` and `\nparentref` are useful for making subitems whose names derive from their parent’s:

```

\DeclareNumberedStyle{inner}[
  counter format=\alph{inner},
  display format=#1.]
\begin{prop}
  \item[OI] Outer item.
  \label{outer2}
  \begin{prop}
    \item[style=inner,
      format=\parentref{.#1}]
    \label{dsub1}
    Ref: \ref{dsub1},
    naked: \nref{dsub1}.
    \item[style=inner,
      display format=#1.,
      ref format=\parentref{.#1}]
    \label{dsub2}
    Ref: \ref{dsub2},
    naked: \nref{dsub2}.
  \end{prop}
\end{prop}

```

OI Outer item.

OI.a Ref: **OI.a**, naked: a.

b. Ref: **OI.b**, naked: b.

The built-in `leveltwo` and `levelthree` styles have `ref format=\parentref{#1}` and `ref format=\parentref{.#1}`, respectively, so that if the parent references as ‘(P1)’, a `leveltwo` sub-item references as ‘(P1a)’ and `\nref` returns just ‘a’.

8.1 How it works: `\propapply`

`\propapply{<template>}{<content>}`

Internally, each reference is stored in the `.aux` file as `\propapply{<template>}{<content>}`.

The `<template>` contains formatting with the placeholder `\propfmtarg` where content appears. At reference time, `\propapply` evaluates the template with `\propfmtarg` bound to `<content>`. The `\oref`^{P.13} and `\nref`^{P.13} commands work by locally redefining `\propapply`.

In normal use, you need not interact with `\propapply` directly.

`\propfmtarg`

Placeholder used inside templates; expands to the content argument of the enclosing `\propapply`.

9 Package options

`\proptoptions{<keys>}`

Sets package-level keys (see [section 6](#)). These can also be set in the optional argument of `prop` or `inlineprop` (local to that environment), or as options to `\usepackage` (global).

10 Compatibility

The propositions package is designed to work with `hyperref`, `cleveref`, and `amsmath`. `amsmath` is required for `\ptag` and the `equations` option. With `cleveref` loaded,

all proposition items are assigned to a default **proposition** “reference type”; the **crefname** key can override this.

Recommended load order:

```
\usepackage{hyperref}  
\usepackage{amsmath} % if using \ptag  
\usepackage{propositions}  
\usepackage{cleveref} % if used
```

11 Known issues

When using `\ptag` with a named counter (e.g. `\ptag[counter=P]`) inside an `amsmath` equation environment, `hyperref` may emit warnings of the form:

```
pdfTeX warning: destination with the same  
identifier (name{equation.N}) has been already  
used, duplicate ignored
```

These warnings are harmless and do not affect the correctness of cross-references.